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# Development of Magnetic Equivalent Circuit method for surface-mounted permanent magnet synchronous motors

*Opracowanie metody ekwiwalentnego obwodu magnetycznego w silnikach synchronicznych z magnesem stałym zamontowanym na powierzchni*

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**Abstract:** This paper proposes a simplified Magnetic Equivalent Circuit (MEC) method for modeling of surface-mounted permanent magnet (SPM) synchronous motors (SPMSMs). The method has constructed a system of equations based on the physical structure of the electric machine, including the teeth, stator yoke, air gap, and permanent magnets. By exploiting the symmetry and periodicity of the magnetic field, the computational domain is simplified, which reduces the number of circuit elements and shortens calculation time. The nonlinear system of equations is solved using a damped Newton's method to ensure stability and convergence. The analytical results are then compared with that obtained from the finite element method (FEM) to show that the error for flux linkage and back-electromotive force is in the range of 5%–7%. However, the cogging torque has lower accuracy due to the approximate modeling of the air gap. The MEC model has demonstrated that a calculation speed of under 1 second, making it suitable for early-stage parameter analysis, design studies, and motor optimization.

**Keywords** Magnetic Equivalent Circuit (MEC) method, Finite Element Method, surface-mounted permanent magnet (SPM) synchronous motors (SPMSMs), Cogging torque, Magnetic flux, Back EMF

**Streszczenie.** W niniejszej pracy zaproponowano metodę uproszczonego ekwiwalentnego obwodu magnetycznego do modelowania powierzchniowo zamontowanego stałego magnesu w silnikach synchronicznych. Przedstawiona metoda zawiera skonstruowanie systemu równań w oparciu o strukturę fizyczną silnika elektrycznego, w tym, uzębienie (?) jarzmo stojana (?), lukę powietrzną, oraz magnesy stałe. Przez zastosowanie symetrii i okresowości pola magnetycznego, uproszczona została procedura obliczeniowa, co redukuje liczbę elementów obwodowych i skracza czas obliczeń. Zastosowano nieliniowy system obliczeń, wykorzystując przybliżoną metodę Newtona w celu zapewnienia stabilności i konwergencji. Następnie, porównano wyniki analityczne z wynikami uzyskanymi na podstawie metody FEM (metoda elementów skończonych?) aby wykazać, że błąd dla strumienia sprzężenia (?) oraz siły elektromotorycznej znajduje się w zakresie 5% – 7%. Jednakże, moment zaczepowy ma mniejszą dokładność z powodu przybliżonego modelowania szczeliny powietrznej. Model MEC wykazał, że szybkość obliczania może być poniżej 1 sek., co czyni go odpowiednim do zastosowania w wczesnej fazie obliczania parametrów, w badaniach projektowych oraz optymalizacji silnika.

**Słowa kluczowe:** metoda ekwiwalentnego obwodu magnetycznego (MEC), metoda elementów skończonych (FEM), magnes stały zamontowany powierzchniowo (SPM), silniki synchroniczne, moment zaczepowy, przepływ magnetyczny, siła elektromotoryczna

## Introduction

Surface-mounted permanent magnet (SPM) motors are increasingly utilized in drive systems due to their high torque density, high efficiency, compact size, and convenient control. These characteristics make SPMs a popular choice for high-performance applica-

tions such as electric vehicles, robotics, and precision industrial equipment [1]. However, the process of designing an optimal SPM motor is complex, often requiring many iterative analysis loops with continuous adjustments to parameters such as dimensions, materials, winding configurations, or air gap distance [2], [3].

Each performance evaluation requires re-simulating the electromagnetic model, leading to long calculation times and high processing costs when using numerical methods like the finite element method. Therefore, in the preliminary design stage, fast-computing models that still ensure acceptable accuracy are essential to support design exploration and optimization [4]. Many studies have proposed the Magnetic Equivalent Circuit (MEC) model for various types of electric motors, showing quite good simulation and experimental results. Nevertheless, most of these works primarily focus on presenting results without detailing the process of building and solving the MEC system of equations. This omission makes it challenging to apply or improve the model, especially in applications that demand flexible and repetitive calculations. In [5], the paper investigated the development of MEC with predictable core losses for PMSMs and introduces novel models that improve core loss prediction accuracy under various load conditions. Furthermore, potential strategies for enhancing these models are proposed and discussed. In [6], the paper proposed an equivalent magnetic circuit model (EMCM) that incorporates end effects and magnetic saturation in both the stator and rotor cores for magnetic field analysis of a conical permanent magnet synchronous motor (CPMSM). In [7], the paper developed an efficient computational approach that achieves a balance between accuracy and computational speed, aiming to improve the performance of permanent magnet motor drives by incorporating core loss effects into equivalent circuit models. Traditional ECMs for PMMs often neglect core losses, leading to inaccuracies in predicting motor behavior, especially under high-speed, high-frequency, or dynamic operating conditions.

The objective of this research is to build a clear and systematic procedure for modeling the MEC for SPM motors, including a complete description of magnetic components such as the air gap, teeth, stator yoke, and magnet region. The model is designed to be general, easy to implement, and convenient to integrate into analysis or design optimization processes. Finally, the model is validated by comparing it with finite element method (FEM) simulation results to evaluate its accuracy and computational efficiency.

## Magnetic Equivalent Circuit (MEC) method

### Model of MEC

The MEC is developed based on the radial cross-section of the PMSM. In this approach, flux tubes (including ferromagnetic regions and air gaps) are repre-

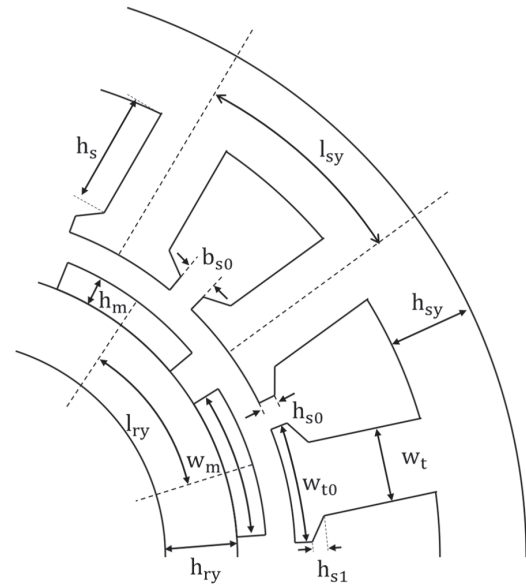


Fig. 1. Cross section of the proposed motor

sented as magnetic reluctances, which are determined by the material permeability and the corresponding geometric dimensions:

$$(1) \quad \mathcal{R} = \frac{l}{\mu_0 \cdot \mu_r \cdot A} (H^{-1}),$$

where  $l(m)$  denotes the length of the flux tube,  $\mu_0 = 4\pi \cdot 10^{-7} (H/m)$  is the permeability of free space,  $\mu_r$  is the relative permeability of the material  $A(m^2)$ ,  $l$  is the cross-sectional area of the flux tube. The reluctance is defined by the dimensional parameters illustrated in Figure 1. The magnetic reluctances are specifically determined as follows:

$$(2) \quad \mathcal{R}_{sy} = \frac{1}{\mu_0 \mu_r} \frac{l_{sy}}{L h_{sy}},$$

$$(3) \quad \mathcal{R}_t = \frac{1}{\mu_0 \mu_r} \frac{h_s + h_{s0} + h_{s1}}{L \overline{wt}},$$

$$(4) \quad \overline{wt} = \frac{w_t h_s + w_{t0} h_{s0} + \frac{h_{s1} (w_{t0} + w_t)}{2}}{h_s + h_{s0} + h_{s1}},$$

$$(5) \quad \mathcal{R}_m = \frac{1}{\mu_0 \mu_r} \frac{h_m}{L l_m},$$

$$(6) \quad \mathcal{R}_{ry} = \frac{1}{\mu_0 \mu_r} \frac{l_{ry}}{L h_{ry}}.$$

The air-gap reluctance elements establish the magnetic connection between the stator and the rotor at a given position. Whenever the rotor passes a stator tooth, the connection in the air gap changes, thereby modifying the topology of the MEC [4]. To maintain a consistent MEC structure throughout the rotor position variation, each permanent magnet (PM) is simultaneously connected to all the remaining teeth [9]. The value of the air-gap reluctance constructed in this manner is defined in Figure 2 [4], [8].

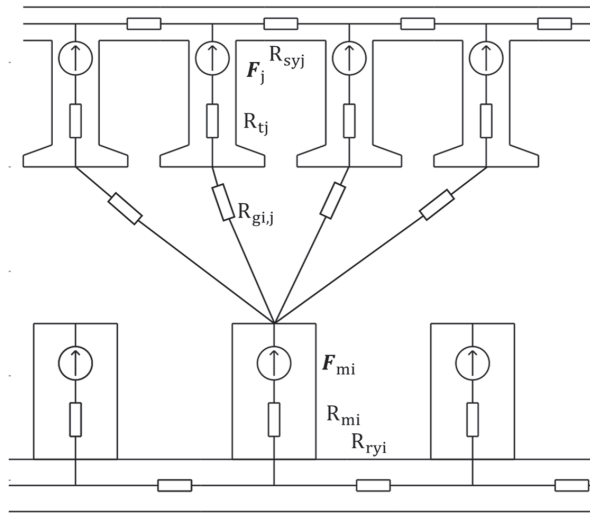


Fig. 2. Equivalent Magnetic Circuit.

$$(7) \quad R_{g,n} = \begin{cases} R_{min}, & 0 \leq \theta_m \leq \theta_{t1} \\ & \text{or } 2\pi - \theta_{t1} \leq \theta_m \leq 2\pi \\ \frac{2R_{min}}{1 + \cos\left(\pi \frac{\theta_m - \theta_{t1}}{\theta_{t2} - \theta_{t1}}\right)}, & \theta_{t1} \leq \theta_m \leq \theta_{t2} \\ \frac{2R_{min}}{1 - \cos\left(\pi \frac{\theta_m - (2\pi - \theta_{t2})}{\theta_m - \theta_{t1}}\right)}, & 2\pi - \theta_{t2} \leq \theta_m \leq 2\pi - \theta_{t1} \\ R_{max}, & \theta_{t2} \leq \theta_m \leq 2\pi - \theta_{t2} \end{cases}$$

where  $l_g(m)$  denotes the air-gap length,  $\mu_g = 1$ , is the overlap area between the stator tooth and the pole,  $\theta_{t1}$  is the angle at which the tooth and pole are still overlapping,  $\theta_{t2}$  is the angle at which the overlap ends  $R_{max}$  and is the disconnection between the tooth and the pole, and it is typically chosen to be sufficiently large to ensure compatibility with computational software.

The MEC is given in Figure 2. It has shown that the winding is placed on the  $j$  stator tooth, and, in the MEC, it is modeled as the magnetomotive force (MMF) source  $F_j(A)$  connected in series with the stator tooth reluctance:

$$(5) \quad F_j = \sum_{k=1}^m N_k \cdot \text{sign}(N_k) \cdot j_k,$$

where  $m$  is the phase number,  $N_k$  is the number of turns of phase  $k$  wound on tooth  $i$ ,  $i_k$  is the current flowing

through the winding of phase  $k$ ,  $\text{sign}(N_k) = -1$  indicates a negative sign if the winding is wound in the opposite direction to the defined convention. sign if the winding is wound in the opposite direction to the defined convention.

The surface-mounted PMs  $m_i$  are represented as MMF sources  $F_{mi}$ . This formulation presents the following expression taking into account the alternating magnet poles:

$$(9) \quad F_{mi} = (-1)^{i-1} \cdot H_0 \cdot l_m(A),$$

where  $H_0(A/m)$  is the coercive force of the PM, is the thickness of the PM.

## 2.1 Reduction of the Computational Domain

According to [10], [11], it is possible to exploit the symmetry and repetitive structure in electrical machine structures to reduce the modeling domain, thereby significantly decreasing the number of elements in the MEC. By only building the model on a representative geometric domain, instead of the entire machine, computation speed is increased while maintaining the accuracy of electromagnetic quantities like linked magnetic flux and back electromotive force (EMF).

The commercial software often applies a technique to reduce the simulation domain based on the simplified slot-to-pole ratio. This is done by taking advantage of the symmetrical and periodic nature of the motor's magnetic field. This approach allows for a significant reduction in the number of geometric elements that need to be simulated, which helps to increase calculation speed while still ensuring the accuracy of electromagnetic quantities as shown in Figure 3.

In this paper, the magnetic circuit is reduced in the number of slots and poles according to the simplified ratio between the number of slots  $Q$  and the number of pole pairs  $p$ .

$$(10) \quad k = \frac{Q}{2p} = \frac{Q'}{2p'}$$

The resistance words that are independent of the rotor's position do not change. In the formula for calculating air gap reluctance in the Equation (7), the angles  $\theta_{t1}$  and  $\theta_{t2}$  are converted based on the original number of slots ( $Q_s$ ) and the number of slots after reduction  $Q'_s$  to be compatible with the reduced MEC.

$$(11) \quad \theta'_{t1} = \theta_{t1} \cdot k$$

$$(12) \quad \theta'_{t2} = \theta_{t2} \cdot k$$

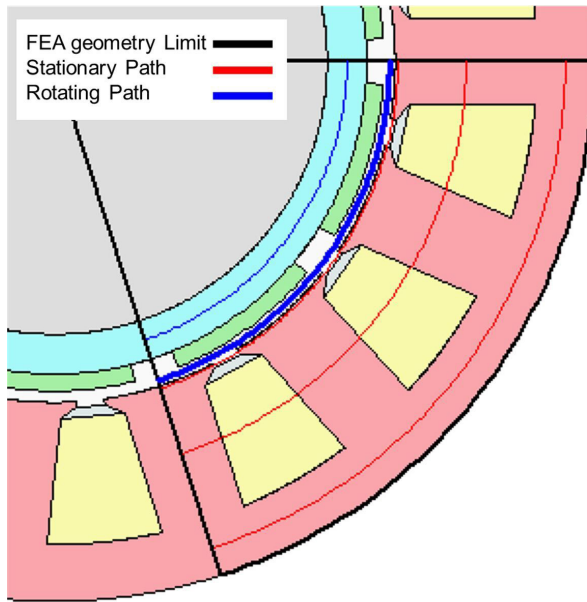


Fig. 3. Reduced research domain.

## Matrix Equation for the MEC

### 3.0 Mathematic Modeling

By neglecting the leakage flux components, the MEC is conducted for a magnetic circuit with a reduced number of slots  $Q'$  and poles  $p'$ . The conventional direction of magnetic flux lines is chosen from left to right, and the loops are defined clockwise. According to the given magnetic circuit in Figure 1, the number of branches in the circuit is

$$(13) \quad m = 2Q' + Q'p' + 2p',$$

The node number is defined as:

$$(14) \quad m = 2Q' + Q'p' + 2p',$$

Number of loop equations is:

$$(15) \quad l = m - n + 1 = Q'p' + 1,$$

The equation of stator yoke can be expressed as

$$(16) \quad \sum_{j=1}^{Q'} R_{syj} \cdot \phi_{syj} = 0,$$

$$(17) \quad \sum_{i=1}^{p'} R_{ryi} \cdot \phi_{ryi} = 0,$$

where  $Q'p' - 1$  is the equation of magnetic flux from stator to rotor:

$$(18) \quad (R_{m,i} \Phi_{m,i} - R_{m,i+1} \Phi_{m,i+1} + R_{gi,j} \Phi_{gi,j} - \\ - R_{gi+1,j+1} \Phi_{gi+1,j+1} + R_{t,j} \Phi_{t,j} - R_{tj+1} \Phi_{tj+1} + \\ + R_{syj} \Phi_{syj} - R_{ryi} \Phi_{ryi}) = F_{m,i} - F_{m,i+1} + F_j - F_{j+1}$$

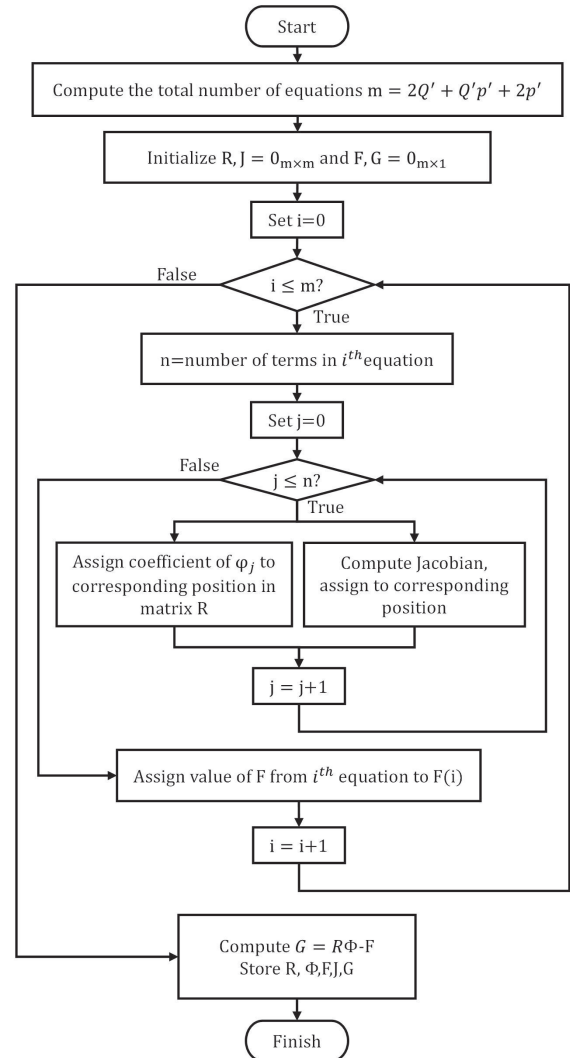


Fig. 4. Algorithm for the system of matrix equations.

The node equation are defined as:

$$(19) \quad k = n - 1 = 2Q' + 2p' - 1,$$

where  $Q'$  are the node equations of stator boundary and air-gap.

$$(20) \quad -\Phi_{syj} + \Phi_{syj-1} + \Phi_{tj} = 0,$$

$$(21) \quad \sum_{i=1}^{p'} \phi_{gi,j} - \phi_{tj} = 0$$

$$(22) \quad -\sum_{j=1}^{Q'} \phi_{gi,j} + \phi_{mi} = 0$$

where  $p' - 1$  are the rotor equation at the boundary:

$$(23) \quad -\Phi_{ryi} + \Phi_{ryi-1} - \Phi_{mi} = 0$$

### 3.1 Jacobian Matrix

The Jacobian matrix is conducted for a reflect matrix, that is  $G = \Phi(\Phi) = F$ , with  $[J_G(\Phi)]_{ij} = \frac{\partial G_i(\Phi)}{\partial \Phi_j}$ :

$$(24) \quad G_i(\Phi) = \sum_{j=1}^m R_{ij}(\Phi) \cdot \Phi_j - F_i = \\ = R_{i,1} \Phi_1 + R_{i,2} \Phi_2 + R_{i,j} \Phi_j + \dots + R_{i,m} \Phi_m - F_i$$

The equations from (1) to (8) present the MEC. It can be seen that  $R_{ij} \Phi_j$  depends on  $\Phi_j$ . By taking the partial derivative of both sides of the  $\Phi_j$ , one gets:

$$(25) \quad \frac{\partial G_i(\Phi)}{\partial \Phi_j} = \frac{\partial}{\partial \Phi_j} (R_{ij}(\Phi_j) \cdot \Phi_j) = \\ = \left( \frac{\partial R_{ij}(\Phi_j)}{\partial \Phi_j} \right) \cdot \Phi_j + R_{ij}(\Phi_j) \cdot \left( \frac{\partial \Phi_j}{\partial \Phi_j} \right) = \\ = \frac{\partial R_{ij}(\Phi_j)}{\partial \Phi_j} \cdot \Phi_j + R_{ij}(\Phi_j)$$

### 3.1 Algorithm for matrix equation

The algorithm for matrix equation defined for next steps is expressed as:

$$(26) \quad R(\Phi) \cdot \Phi = F$$

where  $R \in R^{m \times m}$  is the reluctant matrix defined by the factors of flux from the equations (1) to (8),  $\Phi \in R^{m \times 1}$  is the flux matrix,  $F \in R^{m \times 1}$  is defined from the left side of the equations (1) to (8). Figure 4 presents the way of constructing the matrix.

The MEC system of equations, derived from the equations in Figure above, has the form  $R(\Phi) \cdot \Phi = F$ . To enhance numerical stability and improve convergence during the solution process, the following system of equations using the damped newton method is solved. The damping coefficient,  $(0 < t_k < 1)$ , is adjusted during the search for a solution [14]. The solution algorithm is presented in Figure 5.

## Computation of electromagnetic parameters

### 4.0 Computation of linkage flux

The flux linkage of the phase  $i$   $\Phi_{li}$  is defined as the total flux through the winding  $N$  of phase  $i$ . Thus, it has

$$(27) \quad \Phi_{li} = \sum_{j=1}^Q k N_{i,j} \cdot \text{sign}(N_{i,j}) \cdot \phi_{tj},$$

where  $N_j$  is the number of winding of phase  $i$  on the tooth  $j$ ,  $\text{sign}(N_{i,j}) = -1$ . If the winding is in the reverse direction of the convention  $\Phi_{ij}$ , is the magnetic flux flowing into the  $k$  tooth, which is the simplified ratio of the number of slots before and after reduction.

### 4.1 Back Electromotive Force

The back EMF of phase  $i$  is defined by the time derivative of the linkage flux  $i$ . It should be noted that the speed of rotary magnetic field equals to  $p$  times of the rotational speed of the shaft  $v_s$  (rpm):

$$(28) \quad e_i = -\frac{d\Phi_i}{dt} = -\frac{d\Phi_i}{d\theta} \cdot \frac{d\theta}{dt} = -\frac{d\Phi_i}{d\theta} \cdot \frac{v_s p}{60} \text{ (V)}$$

### 4.2 Analysis of cogging torque

The modeling of cogging torque is a type of no-load torque that occurs in electric motors, especially in PMSMs. It appears even when no current is supplied to the windings, caused by the interaction between the

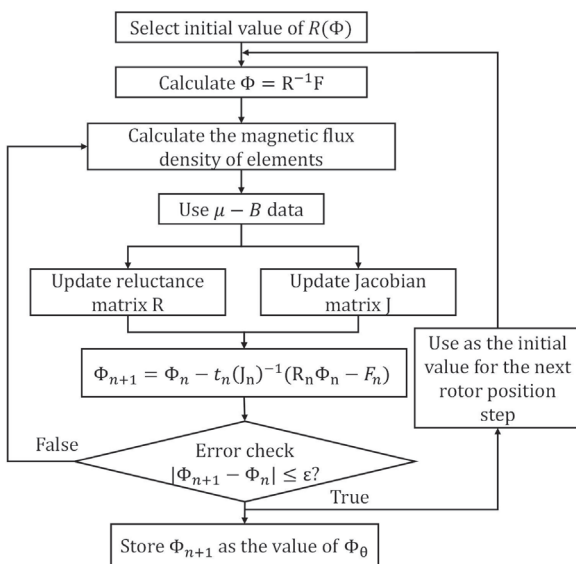


Fig. 5. Algorithm for flux distribution.



rotor's magnetic field and the variation of air-gap permeance as the rotor moves past different positions of the stator teeth, defined in [12], [13].

$$(29) \quad T_{cog}(\theta) = -\frac{dW_m(\theta)}{d\theta},$$

where  $W_m(\theta)$  is the magnetic energy in the air-gap that depends on the rotor angle ( $\theta$ ).

The sign “-” shows the torque tends to move toward the position of minimum energy.

In this study, the leakage flux is neglected. Therefore, the total flux flowing in the air-gap equals the magnet flux. As a result, the magnetic energy can be expressed in terms of the magnet flux ( $\Phi_m(\theta)$ ) and the magnetomotive force of the PM  $F_m(\theta)$ :

$$(30) \quad W_m(\theta) = \frac{1}{2} F_m(\theta) \Phi_m(\theta),$$

In the simplified MEC with magnetomotive force sources, one uses:

$$(31) \quad W_m(\theta) = \frac{k}{2} \sum_{i=1}^{p \setminus r q} \Phi_{mi}(\theta) \cdot F_{mi}(\theta)$$

## Application test

The motor used for calculation and comparison with FEM is the SPMSM with concentrated windings. The detailed configuration parameters are provided in Table 1.

The distribution of magnetic field in a electrical angle  $\Phi(\theta)$  is computed via the Algorithm as already given in Figure 5, where  $\theta$  varies within one electrical angle. Figure 6 presents the computed distribution of the magnetic field (flux distribution) within the magnetic circuit over one electrical angle. This visualization is fundamental for understanding how magnetic flux is distributed across different components of the motor, providing insights into its basic magnetic behavior.

Figure 7 displays the computed results for the flux linkage. It can be seen that the flux linkage is a critical electromagnetic quantity directly related to the motor's performance. When compared with the FEM simulations, the MEC model achieves high accuracy for flux linkage, with an approximate error of 4.5%. Furthermore, the waveforms and variation trends of the magnetic flux show good agreement with FEM results. Figure 8 illustrates the distribution of the back EMF, which is computed at the motor's rated shaft speed. The back-EMF is essential for evaluating the motor's voltage generation characteristics. The MEC model demonstrates good accuracy for back-EMF, with an error of approximately 7% when compared

Table 1. The parameters of the proposed motor

| Parameters                | Value      |
|---------------------------|------------|
| Rated power               | 7.5 kW     |
| Rated speed               | 1500 rpm   |
| Slot number               | 15         |
| Number of pole pair       | 5          |
| Outer diameter stator     | 190 mm     |
| Inner diameter stator     | 120 mm     |
| Tooth width               | 15 mm      |
| Deep slot                 | 25 mm      |
| Slot neck depth           | 1 mm       |
| Open slot                 | 3 mm       |
| PM thickness              | 3 mm       |
| PM opening angle          | 140 degree |
| Length of air gap         | 2 mm       |
| Shaft diameter            | 96 mm      |
| Length of machine         | 120 mm     |
| Turn number of each phase | 29 turns   |

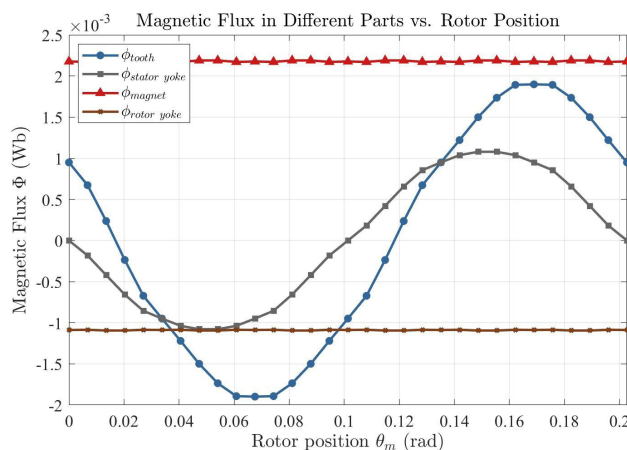


Fig. 6. Flux distribution in the magnetic circuit.

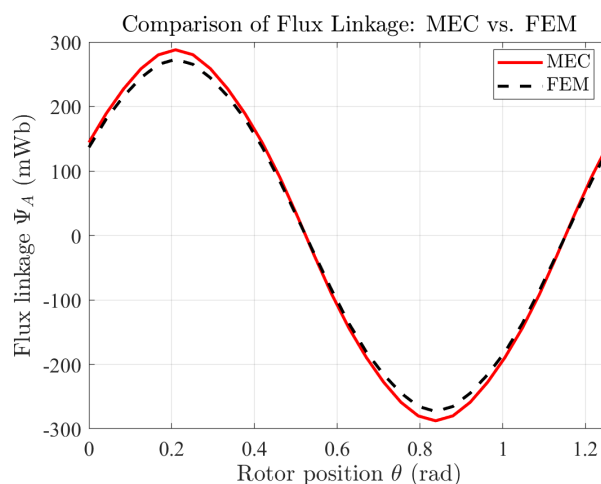


Fig. 7. Flux linkage.

to FEM simulations. The reduced MEC model is considered suitable for predicting induced electromotive force, with its waveforms and trends showing good agreement with FEM results.

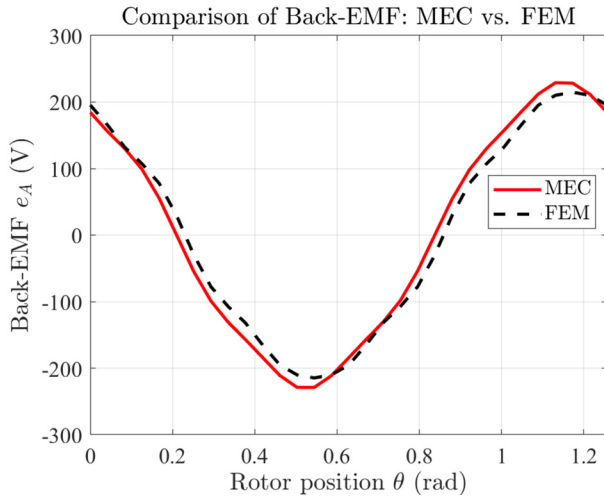


Fig. 8. Distribution of back EMF

Figure 9 shows the distribution of cogging torque as a function of rotor position. the cogging torque is a type of no-load torque in permanent magnet motors, caused by the interaction between the rotor's magnetic field and variations in air-gap permeance. While the MEC model's waveforms and variation trends for cogging torque align well with FEM results, its accuracy for this parameter is lower, exhibiting an error of about 20%. This discrepancy is attributed to the approximate modeling of the air-gap reluctance and the neglect of leakage flux. Consequently, the reduced MEC model is "not yet fully reliable for analyzing small torque ripples".

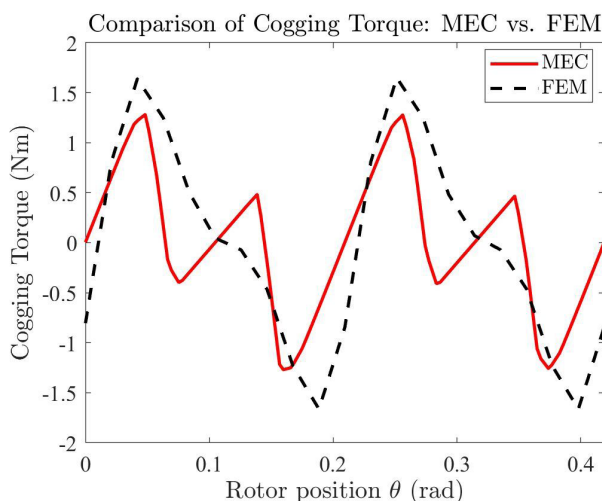


Fig. 9. Distribution of cogging torque according to the position of rotor.

Figure 10 demonstrates a significant practical application of the MEC model in the early design stage: it illustrates how MEC data can be used to eliminate design regions that do not satisfy specific back-EMF requirements. This capability highlights the MEC model's utility in quickly evaluating a large number of design parameters, owing to its very high computational speed (under 1 second per configuration). This rapid evaluation process significantly aids in motor geometry optimization and reduces the computational cost compared to conventional FEM.

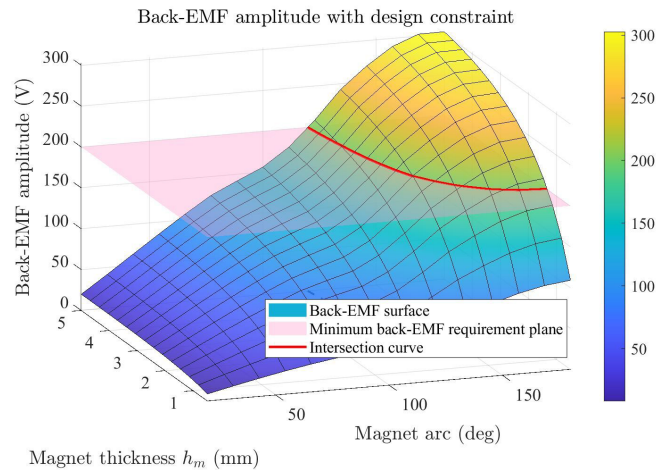


Fig. 10. Using MEC to eliminate designs that do not meet the back-EMF requirements.

Figure 11 showcases another valuable application of the MEC tool: its capability to predict the magnetic circuit saturation state based on design parameters such as tooth width and slot height. Predicting saturation is crucial for designing efficient motors, as saturation can limit performance. Alongside Figure 10, this demonstrates that the MEC tool not only facilitates fast performance evaluation but also extends its applicability to multi-objective optimization, thereby significantly reducing computational cost compared to conventional FEM methods.

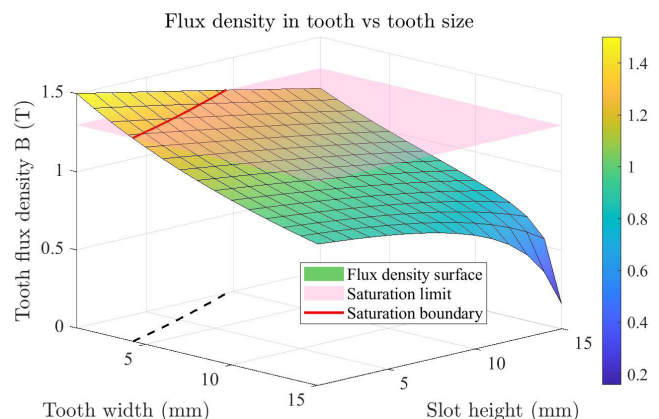


Fig. 11. The saturation state of the magnetic circuit is predicted using MEC data.

The simulation results, compared with the FEM, show that the proposed MEC model achieves high accuracy for most electromagnetic quantities. The calculation error of the flux linkage is approximately 4.5%, and that of the back-EMF is about 7%, while the cogging torque error is larger (around 20%) due to the approximation of the air-gap reluctance and the neglect of leakage flux. This indicates that the reduced MEC model is suitable for predicting flux distribution and induced EMF, but is not yet fully reliable for analyzing small torque ripples.

The outstanding advantage of the method lies in its very high computational speed (less than 1 second per configuration), achieved by reducing the number of elements in the magnetic circuit and omitting components with minor influence. This enables rapid exploration of a large number of design parameters, thereby supporting the motor geometry optimization process in the early design stage.

## Conclusion

This paper presents a procedure for constructing a reduced-order MEC for a SPMSM. The modeling approach includes detailed representations of the magnetic reluctances in the stator yoke, teeth, magnets, and air gap, as well as an effective method for reduc-

ing the computational domain by exploiting the periodicity of the magnetic field. The MEC system of equations is represented in matrix form and solved using the Damped Newton Method, which ensures stability and convergence. During the modeling process, leakage flux was neglected, which led to an amplitude error of up to approximately 10% compared to FEM simulations. However, the waveforms and variation trends of electromagnetic quantities such as magnetic flux, back EMF, and cogging torque still showed good agreement with the FEM results. Thanks to the reduction in the number of magnetic circuit elements, the model's computational time was under 1 second. This makes it suitable for problems that require multiple iterations during the preliminary design or optimization phase.

The proposed method exhibits great potential for practical application in the design of surface-mounted permanent magnet (SPM) motors. Its flexible formulation allows extension to various motor topologies and integration with optimization algorithms to enhance design efficiency, while substantially reducing the dependence on time-consuming FEM simulations during the preliminary design stage. Consequently, the reduced MEC model serves as a robust and computationally efficient framework for early-stage motor analysis and optimization.

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